

Finite-Model Kill Tests for a Gap–Cone Dependency Chain and the 4/3 Coincidence

Diagnostic note accompanying ACS Framework Issue #7

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with computational verification in-repo

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Abstract

Two diagnostic programmes are reported as finite-model kill tests, not as theorems. The first asks whether a proposed dependency direction

spectral gap $\rightarrow$ clustering $\rightarrow$ cone-sharpness proxy

is supported as a robust monotone relation on small lattice models (TFIM, XXZ+ h , long-range Ising) and on an exact float-free classical Ising analogue. The second asks whether the numeric coincidence between an ACS-side constant $\beta = 4/3$ and a density-side exponent form $\alpha(d) = 1 + 1/d$ at $d = 3$ survives mechanism-level robustness checks. Across the tested windows the Section 9 chain is not supported as a robust monotone relation; the 4/3 equality fails normalisation robustness and collapses to the identity family $n = d + 1$ on the scanned integer lattice. All scripts, JSON artefacts, SHA-256 anchors, and an end-to-end verification pipeline are included in-repo. Claims are scoped to tested models and parameter windows.

Publication status (24 September 2026). This version includes a claim-specific amendment at the end of the main text. That amendment governs earlier physical interpretations and distinguishes retained calculations from withdrawn or unresolved inferences.

License

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1 Scope and claim discipline

This note reports finite tested outcomes only. It does not claim a universal no-go theorem, a proof of cone rigidity, or a derivation of a critical coupling. All findings are bounded by the tested model families, parameter windows, and (where applicable) numeric backend.

Remark 1 (Relation to Little-type programmes). *A dependency statement of the form “static reduction is meaningful only after observables are rigid enough to support a sharp cone” motivates the Section 9 toy, but the present work does not establish that dependency as a theorem. The tests ask whether a computable proxy of the proposed direction survives kill criteria on finite models.*

2 Research questions

RQ-1 (Section 9 toy chain). In finite lattice models, does larger spectral gap co-vary with both shorter static clustering length and lower outside-cone commutator leakage (or an exact influence analogue)?

RQ-2 (4/3 mechanism). Is the equality at 4/3 generated by a robust mechanism, or does it collapse under normalisation changes and bookkeeping-identity checks?

3 Methods

3.1 Section 9 float numerics

Scripts under `code/issue7/`:

- `section9_toy_kill_test.py` — baseline TFIM toy;
- `exhaustive_issue7.py` — multi-size / multi-threshold sweep;
- `cross_model_kill_pass.py` — TFIM and XXZ+ h with dual leakage summaries;
- `long_range_kill_pass.py` — power-law Ising, $\alpha \in \{3, 2\}$.

Observables: gap $\Delta = E_1 - E_0$; clustering length ξ from connected Z_0 - Z_r correlators; outside-cone leakage from $\| [Z_0(t), Z_r] \|$ (max and, where stated, mean).

Decision rule (float suite): support requires $\text{corr}(\text{gap}, \xi) < -0.2$ and $\text{corr}(\text{gap}, \text{leak_proxy}) < -0.2$ for every required leakage proxy.

3.2 Section 9 exact (no IEEE float)

`section9_exact_kill_test.py` uses integer Ising energies, integer Boltzmann base B , connected correlators in `Fraction`, and local-majority influence fractions in `Fraction`. The decision path uses only signs of `Fraction` covariances.

Companion exact ACS TE automaton (already in-repo): `code/acs_codebase/extras/integer_acs.py`.

3.3 4/3 mechanism tests

`mechanism_4over3_test.py` and the 4/3 half of `exhaustive_issue7.py`:

1. K1 — normalisation robustness: scale $\beta = 4/3$ by c and inspect $\hat{d} = 1/(c\beta - 1)$;
2. K2 — integer lattice scan of $\alpha(d) = 1 + 1/d$ against $\beta(n) = n/(n - 1)$.

3.4 End-to-end verification

`verify_issue7_pipeline.py` reruns all Issue #7 scripts, rewrites canonical stdout logs, checks stdout-vs-JSON consistency, and verifies SHA-256 anchors listed in the companion markdown artefacts under `docs/`.

4 Results

4.1 Section 9 baseline and exhaustive

Baseline TFIM ($N = 7$): $\text{corr}(\text{gap}, \xi) = -0.6375$, $\text{corr}(\text{gap}, \text{leak}) = +0.9636$ — chain not supported.

Exhaustive float sweep: support rate 0/12. Gap- ξ correlations remain negative; gap-outside-leak correlations remain positive in every tested cell.

4.2 Cross-model and long-range

Family	Support	Rate
TFIM (cross-model grid)	0/6	0.00
XXZ+ h	1/6	0.17
Long-range $\alpha = 3$	2/6	0.33
Long-range $\alpha = 2$	3/6	0.50

The single XXZ support cell and the long-range support cells concentrate at the smallest leakage threshold (10^{-3}). Partial support is threshold-dependent, not a uniform monotone trend. Long-range Ising — where strict Lieb–Robinson cones are known to weaken — shows the highest partial-support fraction, consistent with threshold sensitivity of the cone proxy rather than establishment of the chain.

4.3 Exact float-free Section 9

- $\text{sign cov}(\text{gap}, \text{cluster_weight}) = -1$;
- $\text{sign cov}(\text{gap}, \text{outside_affect}) = 0$ (outside-cone affect identically zero under local majority);
- joint chain support: false.

`integer_acs.py` on $\mathbb{Z}_{16}$: uncoupled/symmetric $\Delta I \approx 0$; asymmetric $\Delta I \approx \pm 1.499$ with sign reversal under $f \leftrightarrow g$. Probabilities are exact rational.

4.4 4/3 mechanism

Baseline $\beta = 4/3$ implies $\hat{d} = 3$. Under normalisation scaling, inferred-dimension spread is 74 on the tested grid, with non-finite values on part of the interval. Exact lattice hits all satisfy $n = d + 1$. Mechanism not established on this surface.

4.5 Verification

End-to-end pipeline status: `PASS_WITH_CAUTION`; all scripts exit successfully; consistency errors 0; documentation hash checks 0.

5 Reproducibility

Canonical artefacts and SHA-256 anchors are listed in `docs/issue7_paper_section9_and_4over3.md` and `docs/issue7_takeup_section9_and_4over3.md`.

Observed environment for float suite and anchors: Python 3.14.6, NumPy 2.5.0, macOS-15.3.2-arm64-arm-64bit-Mach-0.

Listing 1: Reproduction commands

```
python3 code/issue7/section9_toy_kill_test.py
python3 code/issue7/mechanism_4over3_test.py
python3 code/issue7/exhaustive_issue7.py
python3 code/issue7/cross_model_kill_pass.py
python3 code/issue7/long_range_kill_pass.py
python3 code/issue7/section9_exact_kill_test.py
python3 code/issue7/verify_issue7_pipeline.py
```

6 Limitations

- Float NumPy Section 9 scripts are byte-anchored to a single platform; their *verdicts* are now cross-platform confirmed.¹
- Exact Section 9 / integer ACS diagnostics are float-free on the decision path.
- Finite-size and finite-window coverage can miss alternate regimes.
- No claim is made beyond tested models and tested parameter windows.

7 Conclusion

Within the reported finite tests, the Section 9 dependency chain is not supported as a robust monotone relation across the tested model families and thresholds. The long-range family shows the highest partial-support fraction (5/12), concentrated at the smallest leakage threshold — a threshold-dependence effect, not evidence of a stable universal trend. The exact float-free analogue also fails the joint chain criterion (cluster covariance sign correct; outside-cone affect identically zero under local majority). The 4/3 equality does not satisfy the tested mechanism robustness criteria and is consistent with an identity-family explanation on the scanned lattice. The package is reproducible in-repo with anchored artefacts; end-to-end verification retains caution only for the remaining float NumPy Section 9 scripts.

¹Float NumPy Section 9 scripts (exact diagonalisation / commutator norms) are byte-anchored to one machine (macOS 15.3.2 arm64, Python 3.14.6, NumPy 2.5.0), with same-machine byte-identical reruns recorded in `docs/issue7_verification_report.json`. A second-platform replication (Linux x86_64, Python 3.11, NumPy 2.4; `docs/issue7_linux_replication/`) reproduces every decision-level output exactly (all `supports_chain` booleans and support rates identical; float values agree to $\sim 10^{-12}$), and the exact-arithmetic artifact is byte-identical across the two platforms — the tolerance-based cross-platform anchor this limitation called for. The exact Section 9 kill test and `integer_acs.py` use integer/`Fraction` on the decision path; for those diagnostics, cross-machine IEEE float drift is not a concern and a second platform is not required. The 4/3 lattice/identity checks are likewise exact.

Tier map (MANIFEST)

Claim	Tier	Evidence
TFIM exhaustive chain support rate 0/12	T1	<code>exhaustive_issue7.py</code>
Cross-model combined support 1/12	T1	<code>cross_model_kill_pass.py</code>
Long-range combined support 5/12	T1	<code>long_range_kill_pass.py</code>
Exact joint chain support false	T1	<code>section9_exact_kill_test.py</code>
4/3 identity family $n = d + 1$	T1	<code>mechanism_4over3_test.py</code>
4/3 mechanism established	T4	failed K1/K2
Section 9 chain as general theorem	—	not claimed

Amendment: scope and evidence, 24 September 2026

The kill tests constrain the particular cone chain and normalization hypotheses that were tested. A surviving algebraic coefficient does not select an action normalization or an observed coupling. The later scalar audit retains this distinction with explicit kinetic metrics and alternative complete potentials sharing neutral data. Earlier falsifications remain falsifications; numerical agreement in another channel does not reinstate them.

For the complete-action and microscopic-carrier audit, see the companion papers *Complete Scalar Actions and Identifiability Boundaries in the ACS Framework* and *Condensate Binding, Charge Protection, and the Boundary of Particle Identification* in this repository’s `papers/updates/` directory. The dated publication record separates fresh portable checks from preserved archival results; it does not claim a complete rerun of every historical experiment or a proof of RH.